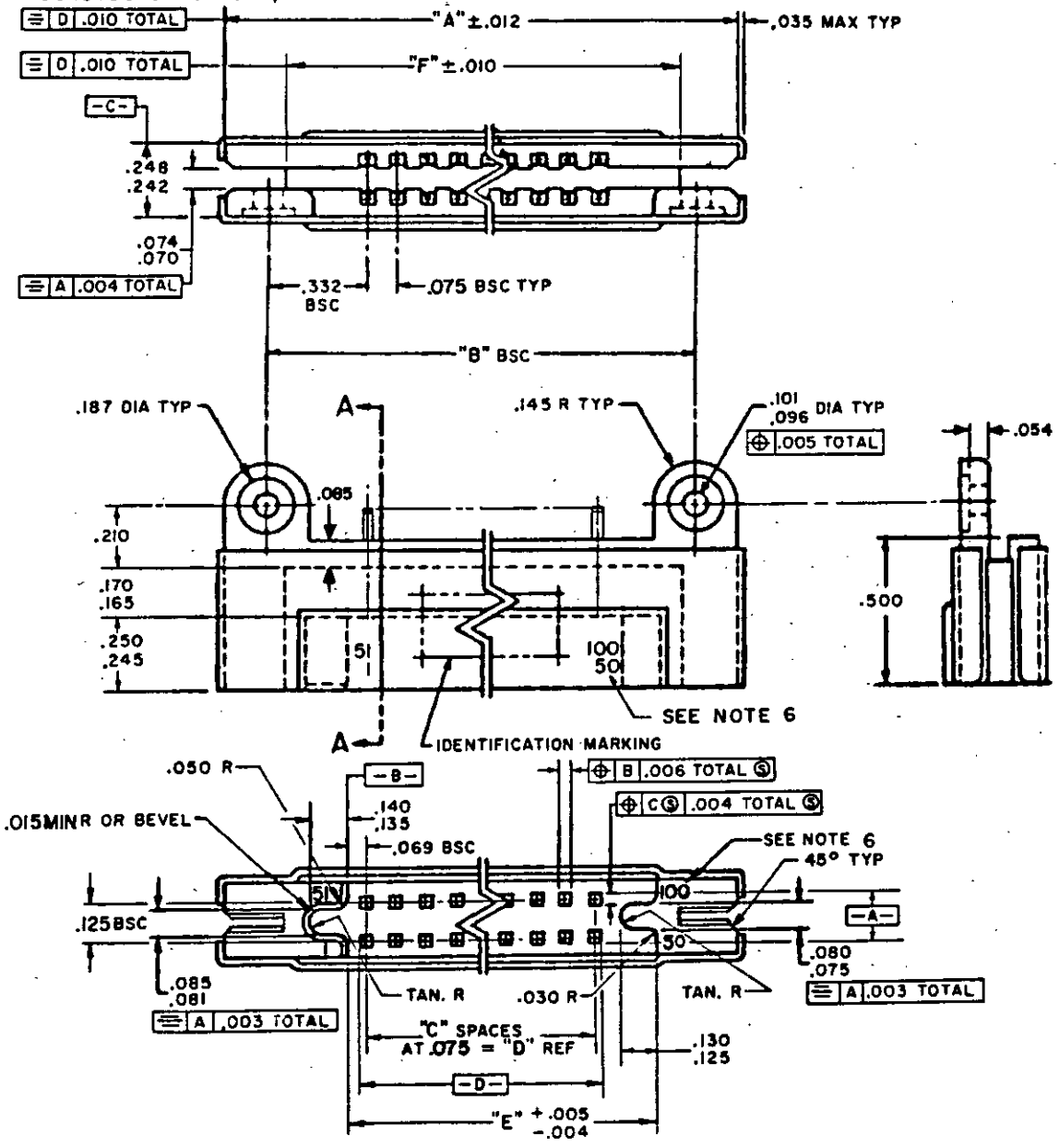
INCH-POUND
MIL-C-55302/23F
31 October 1988
SUPERSEDING
MIL-C-55302/23E
24 March 1980

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, DECADE INCREMENT 10 THRU 120, 150, 160, AND 180 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.075 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.



STYLE I (INTERNAL KEYING)

FIGURE 1. Connectors, plug, (.075 spacing).

Ⓕ denotes changes

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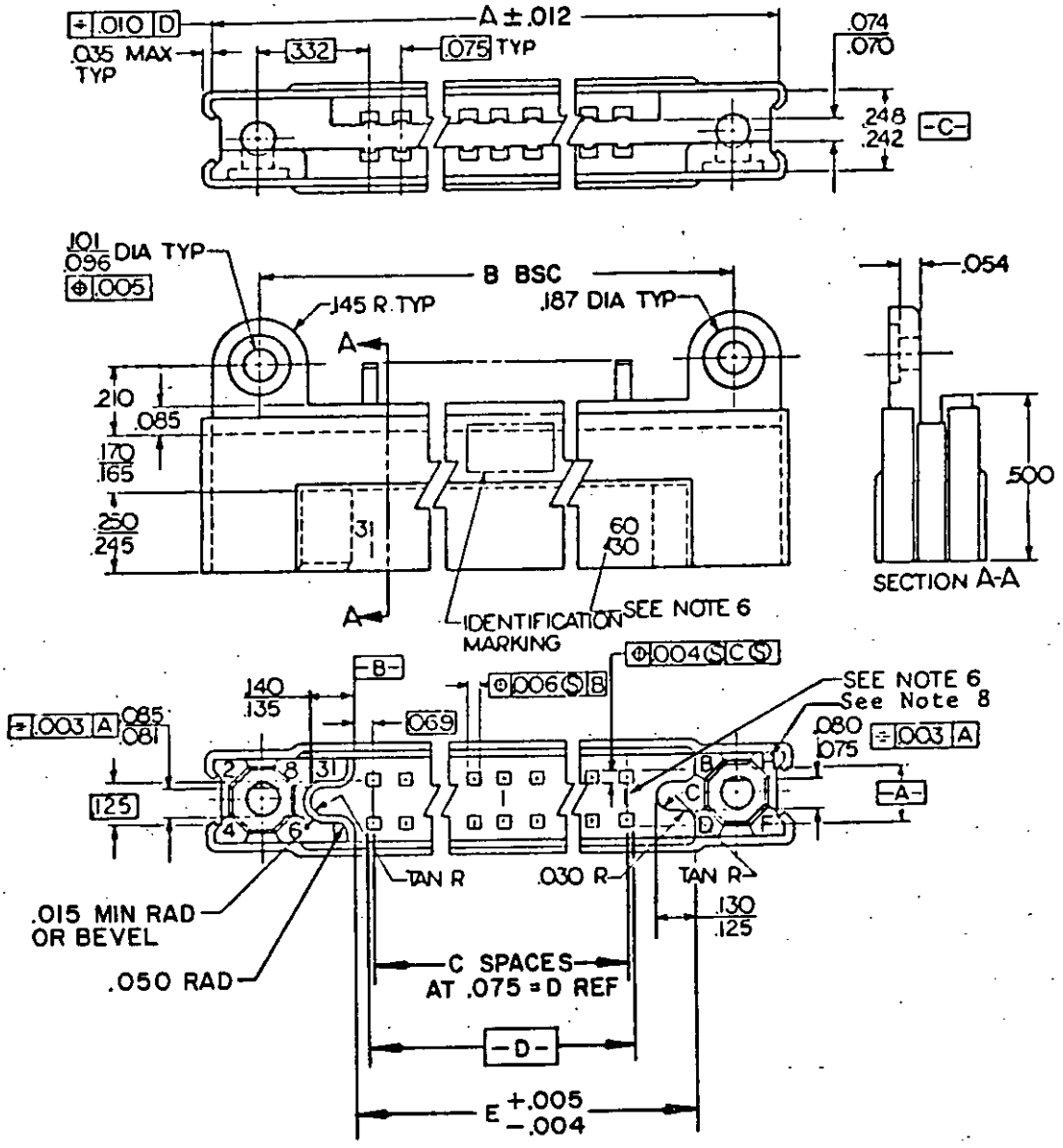
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1/ Dash number	Number of contacts	Dimensions						Contact identification numbers
		A	B	C	D	E	F	
01	10	1.255 (31.88)	.965 (24.51)	4	.300 (7.62)	.575 (14.61)	.855 (21.72)	1 - 5 6 - 10
02	20	1.630 (41.40)	1.340 (34.04)	9	.675 (17.15)	.950 (24.13)	1.230 (31.24)	1 - 10 11 - 20
03	30	2.005 (50.93)	1.715 (43.56)	14	1.050 (26.67)	1.325 (33.66)	1.605 (40.77)	1 - 15 16 - 30
04	40	2.380 (60.45)	2.090 (53.09)	19	1.425 (36.20)	1.700 (43.18)	1.980 (50.29)	1 - 20 21 - 40
05	50	2.755 (69.98)	2.465 (62.61)	24	1.800 (45.72)	2.075 (52.71)	2.355 (59.82)	1 - 25 26 - 50
06	60	3.130 (79.50)	2.840 (72.14)	29	2.175 (55.25)	2.450 (62.23)	2.730 (69.34)	1 - 30 31 - 60
07	70	3.505 (89.03)	3.215 (81.66)	34	2.550 (64.77)	2.825 (71.76)	3.105 (78.87)	1 - 35 36 - 70
08	80	3.880 (98.55)	3.590 (91.19)	39	2.925 (74.30)	3.200 (81.28)	3.480 (88.39)	1 - 40 41 - 80
09	90	4.255 (108.08)	3.965 (100.71)	44	3.300 (83.82)	3.575 (90.81)	3.855 (97.92)	1 - 45 46 - 90
10	100	4.630 (117.60)	4.340 (110.24)	49	3.675 (93.35)	3.950 (100.33)	4.230 (107.44)	1 - 50 51 - 100
11	110	5.005 (127.13)	4.715 (119.76)	54	4.050 (102.87)	4.325 (109.86)	4.605 (116.97)	1 - 55 56 - 110
12	120	5.380 (136.65)	5.090 (129.29)	59	4.425 (112.40)	4.700 (119.38)	4.980 (126.49)	1 - 60 61 - 120
13	150	6.505 (165.23)	6.215 (157.86)	74	5.550 (140.97)	5.825 (147.96)	6.105 (155.07)	1 - 75 76 - 150
14	160	6.880 (174.75)	6.590 (167.39)	79	5.925 (150.50)	6.200 (157.48)	6.480 (164.59)	1 - 80 81 - 160
15	180	7.630 (193.80)	7.340 (186.44)	89	6.675 (169.55)	6.950 (176.53)	7.230 (183.64)	1 - 90 91 - 180

1/ Dash numbers 01 through 15 are for style 1 (internal keying) connectors.

FIGURE 1. Connectors, plug, (.075 spacing) - Continued.

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STYLE 2 (EXTERNAL KEYING)

FIGURE 1. Connectors, plug, (.075 spacing).

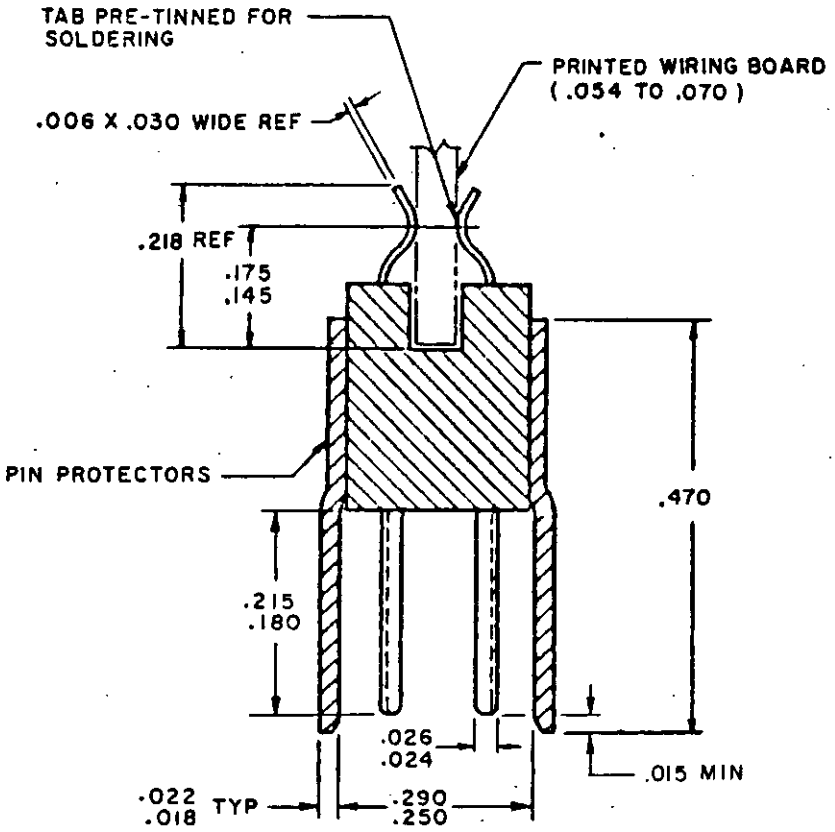
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1/ Dash number	Number of contacts	Dimensions					Contact identification numbers
		A	B	C	D	E	
16	10	1.255 (31.88)	.965 (24.51)	4	.300 (7.62)	.575 (14.61)	1 - 5 6 - 10
17	20	1.630 (41.40)	1.340 (34.04)	9	.675 (17.15)	.950 (24.13)	1 - 10 11 - 20
18	30	2.005 (50.93)	1.715 (43.56)	14	1.050 (26.67)	1.325 (33.66)	1 - 15 16 - 30
19	40	2.380 (60.45)	2.090 (53.09)	19	1.425 (36.20)	1.700 (43.18)	1 - 20 21 - 40
20	50	2.755 (69.98)	2.465 (62.61)	24	1.800 (45.72)	2.075 (52.71)	1 - 25 26 - 50
21	60	3.130 (79.50)	2.840 (72.14)	29	2.175 (55.25)	2.450 (62.23)	1 - 30 31 - 60
22	70	3.505 (89.03)	3.215 (81.66)	34	2.550 (64.77)	2.825 (71.76)	1 - 35 36 - 70
23	80	3.880 (98.55)	3.590 (91.19)	39	2.925 (74.30)	3.200 (81.28)	1 - 40 41 - 80
24	90	4.255 (108.08)	3.965 (100.71)	44	3.300 (83.82)	3.575 (90.81)	1 - 45 46 - 90
25	100	4.630 (117.60)	4.340 (110.24)	49	3.675 (93.35)	3.950 (100.33)	1 - 50 51 - 100
26	110	5.005 (127.13)	4.715 (119.76)	54	4.050 (102.87)	4.325 (109.86)	1 - 55 56 - 110
27	120	5.380 (136.65)	5.090 (129.29)	59	4.425 (112.40)	4.700 (119.38)	1 - 60 61 - 120
28	150	6.505 (165.23)	6.215 (157.86)	74	5.550 (140.97)	5.825 (147.96)	1 - 75 76 - 150
29	160	6.880 (174.75)	6.590 (167.39)	79	5.925 (150.50)	6.200 (157.48)	1 - 80 81 - 160
30	180	7.630 (193.80)	7.340 (186.44)	89	6.675 (169.55)	6.950 (176.53)	1 - 90 91 - 180

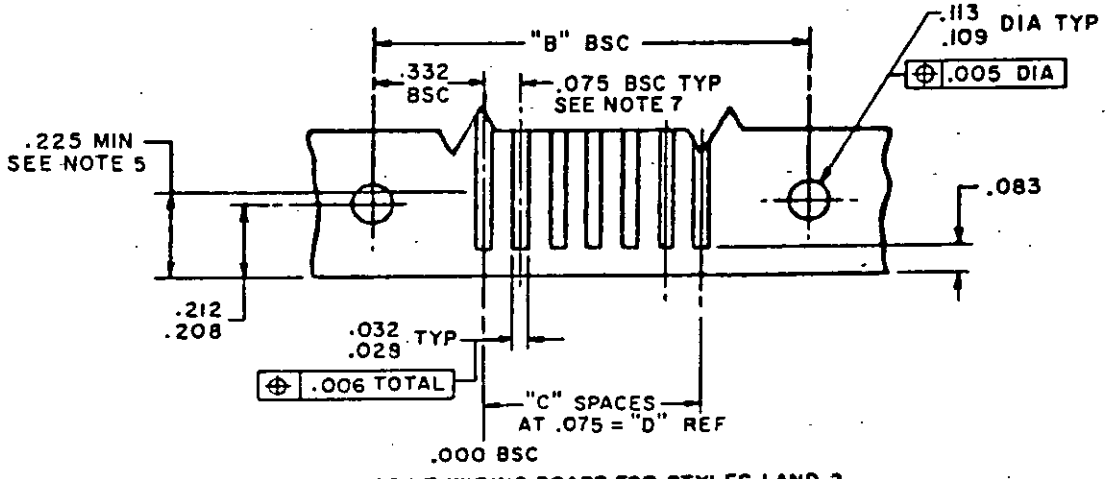
1/ Dash numbers 16 through 30 are for style 2 (external keying) connectors.

FIGURE 1. Connectors, plug, (.075 spacing) - Continued.

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SECTION A-A



RECOMMENDED PRINT WIRING BOARD FOR STYLES 1 AND 2

Figure 1. Connectors, plug, (.075 spacing) - Continued.

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Inches	mm	Inches	mm
.003	.08	.101	2.57
.004	.10	.109	2.77
.005	.13	.113	2.87
.006	.15	.125	3.18
.008	.20	.130	3.30
.010	.25	.135	3.43
.012	.30	.140	3.56
.015	.38	.145	3.68
.018	.46	.165	4.19
.022	.56	.170	4.32
.024	.61	.175	4.44
.026	.66	.180	4.57
.028	.71	.187	4.75
.030	.76	.208	5.28
.032	.81	.210	5.33
.035	.89	.212	5.38
.050	1.27	.215	5.46
.054	1.37	.218	5.54
.069	1.75	.225	5.72
.070	1.78	.242	6.15
.074	1.88	.245	6.22
.075	1.90	.248	6.30
.080	2.03	.250	6.35
.081	2.06	.290	7.37
.083	2.11	.332	8.43
.085	2.16	.470	11.94
.096	2.44	.500	12.70

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/24 or MIL-C-55302/25
5. Minimum pad length on printed circuit board.
6. Numbers indicating, every 5 cavities stamped on side. Numbers indicating end cavities and lines indicating every 5 cavities stamped on mating face.
7. Termination layout on .025 (.64. mm) modular grid.
8. Locations indicators embossed on surface.

FIGURE 1. Connectors, plug, (.075 spacing) - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Ⓕ Material:
Plating: The contact plating for the engagement area, .150 inch minimum length, shall be gold per MIL-C-55302 over nickel per MIL-C-55302. The contact plating for the solder tail area, .160 inch minimum length, shall be tin lead over nickel, with a tin lead dip per MIL-C-55302 for .080 minimum length. The remainder of the contact shall be nickel plated per MIL-C-55302.

Contact identification: See figure 1.

Ⓕ Coating: The aluminum pin protector shall be coated per MIL-A-8625, type II or epoxy coated .001 to .003 inch thick.

Contact rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient.

Keying: See MIL-C-55302/31.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .25 and the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: The average contact resistance of all contacts measured shall not exceed .015 ohm, and no individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: 3 pounds minimum.

Dielectric withstanding voltage:
Sea level: 900 volts rms.
High altitude: 200 volts rms.

Part number: M55302/23- (dash number from figure 1).

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - CR, AR, MI
Navy - SH, AS, OS
Air Force - 11, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3645-01)

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